Events

Source:

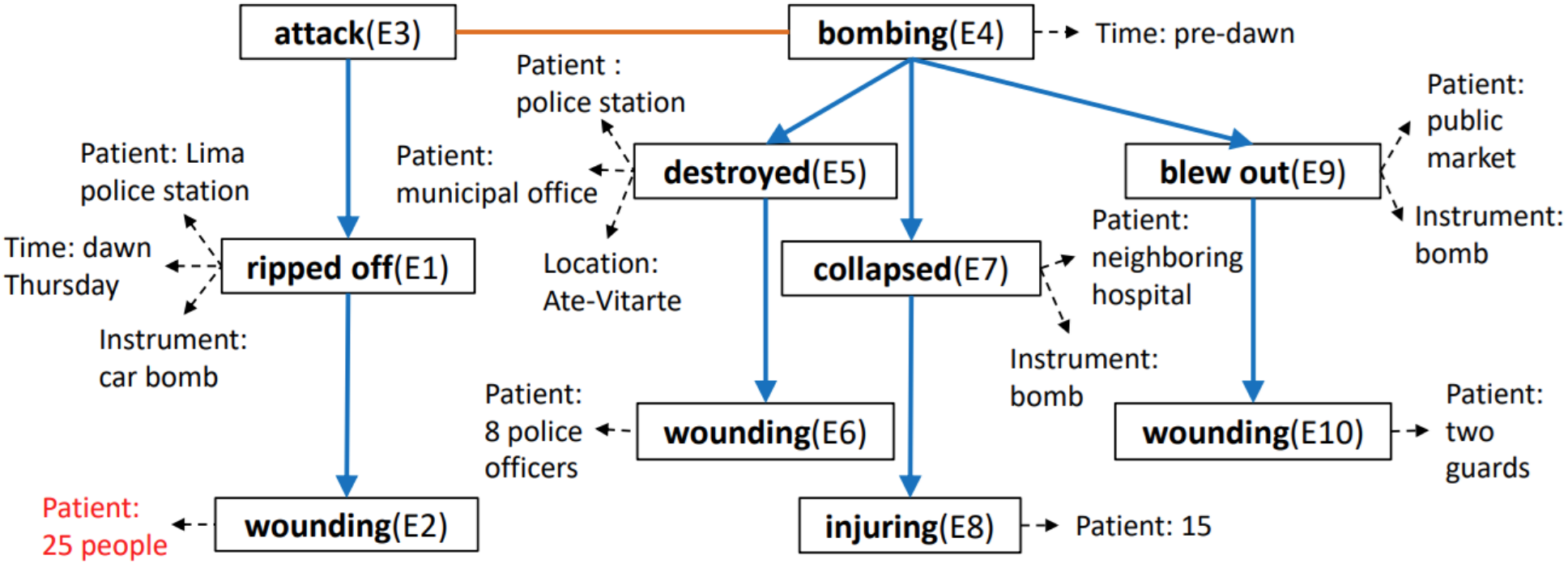
1. Jun Araki. Extraction of Event Structures from Text. Ph.D. Thesis, Carnegie Mellon University. 2018.

Event-oriented text analysis

- Need
- Challenges
  - Defining events
  - Relations between events
- Applications
  - Summarization
  - Question answering
  - Knowledge base population

Events are Complex

A car bomb that police said was set by Shining Path guerrillas **ripped off**(E1) the front of a Lima police station before dawn Thursday, **wounding**(E2) 25 people. The **attack**(E3) marked the return to the spotlight of the feared Maoist group, recently overshadowed by a smaller rival band of rebels. The pre-dawn **bombing**(E4) **destroyed**(E5) part of the police station and a municipal office in Lima's industrial suburb of Ate-Vitarte, **wounding**(E6) 8 police officers, one seriously, Interior Minister Cesar Saucedo told reporters. The bomb **collapsed**(E7) the roof of a neighboring hospital, **injuring**(E8) 15, and **blew out**(E9) windows and doors in a public market, **wounding**(E10) two guards.



https://junaraki.net/thesis/phd-thesis-slides.pdf

Defining and annotating events

- actions, processes, and states.
  - actions that involve a change of state with an explicit goal or completion, e.g., walked to Boston, buy a book;
  - processes that involve a change of state without an explicit goal or completion, e.g., it was raining yesterday;
  - states that remain unchanged until their change or are brought as a result of an event, e.g., He owns a car. Tom was happy when he received a present.

Qualities of Events

Quality	Definition
Occurrence	The quality of a predicate that signals that a change in the state of the world has taken place.
Spatio-Temporal Grounding	The degree to which the predicate is grounded in time and space.
Lexical Aspect	The boundedness and duration of the predicate with respect to time.
Agency	The degree to which the main event participant can be regarded as a “causer” or “doer”.
Affectedness	The degree to which the action of the predicate affects the semantic patient.
Actuality	The knowledge of whether the predicate actually took/takes place.
Specificity	The degree to which the predicate refers to a particular instance of an event.

https://aclanthology.org/W14-2909.pdf

Occurrence

1. He was running on the track. (high energy)

2. He was sitting in the chair. (low energy)

Spatio-Temporal Grounding

1. He fought the law.

2. He fought the law yesterday.

3. He fought the law yesterday in court

Lexical Aspect

1. He is building a house. (telic, durative)
2. He is swimming. (atelic, durative)
3. He shot the man. (telic, punctual)
4. He is knocking on the door. (atelic, punctual)

Agency

1. The wine aged in a barrel. (no agent)

2. The vintner aged the wine in a barrel. (agent: vintner)

Affectedness

1. He punched some pillows. (low individuation)
2. He punched his brother. (high individuation)
3. He bruised the man’s leg. (low affectedness)
4. He broke the man’s leg. (high affectedness)

Specificity

1. The chicken laid an egg on Tuesday. (specific)
2. The chicken lays two eggs a week. (habitual)
3. Chickens lay eggs when fertile. (generic)

Actuality

1. He bought a new car. (realis)
2. He might buy a new car. (future)
3. He might have bought a new car. (epistemic)
4. He is able to buy a new car. (ability)
5. He wants to buy a new car. (deontic)
6. He was prevented from buying a new car. (negative, prevention)
7. He did not buy a new car. (negative, simple negation)

Event Nuggets

- a semantically meaningful unit that expresses the event in a sentence.

  An event nugget can be either a single word (main verb, noun, adjective, adverb) or a continuous or discontinuous multi-word phrase

- Examples:
  - He **shot** the teller in the bank.
  - He **<u>opened fire</u>** at the teller in the bank.
  - He **<u>turned</u>** the television **<u>on</u>**.
  - **Well-known** researchers **<u>gave</u>** a **<u>speech</u>** at the **conference**.

Event Relations

- Event coreference
- Subevent
- Causality
- Subsequence
- Order of events (before/after/simultaneous)

Accenture News Example

- **The acquisition spree** continues from 2019 and the firm has bought a total of 63 smaller companies...
- Accenture CEO Julie Sweet even **went *on to say*** that the new unit is a "once in a digital era re-platforming of businesses" during the company's second quarter earnings call in March this year.
- **The acquisition** of software technology companies Wolox, Imaginea, Cygni and Infinity Works all point toward Accenture's **aggressive push into the cloud**.
- Mountain View, California based cloud engineering services provider Imaginea is Accenture's biggest **buy** this year…
- Joinville, Brazil headquartered Pollux is significant as it is the company's **first acquisition** of an industrial robotics solutions provider. Pollux **produces** real industrial robots that are currently used in 150 locations in Brazil.

https://www.techcircle.in/2021/04/05/accenture-is-on-an-acquisition-spree-around-the-globe-here-s-why

Annotating Eventive Verbs

- Psychological or cognitive verbs
  - I **found** it interesting; They **figured out** the problem
- Aspectual verbs
  - The ice has **begun** to **melt**.
  - In the **beginning**, the Soviet authorities targeted the elimination of illiteracy.
- Causative verbs (let, allow, permit, make, force, require, get, have, help, and keep)
  - He **forced** me **to cancel** the flight.
- Performance verbs
  - She **failed** to **come back** by 7 o'clock. (?)
  - We had to **solve** the problem. So we **did it**.

Annotating Eventive Verbs

- Bridging or supporting verbs
  - He **tried** to **persuade** his parents. (?)
- Communication verbs
  - He was **named** after his father.
- Phrasal verbs
  - She has always **looked down on** me.
- Stative verbs
  - Everybody **liked** her.
  - He **lives** in Chicago for most of his life.
  - The United States **have** 50 states.
- Occurrence verbs
  - The bombing **occurred** last Wednesday.

Light Verbs

- **break** new ground
- **come** clean
- **do** business
- **get** moving
- **give** up
- **had**
- **keep** going
- **make** a scene
- **open** the attack
- **pay** attention
- **set** foot
- **take** an action

https://en.wiktionary.org/wiki/Category:English_light_verb_constructions

Light verb constructions in Hindi

- raam=ne mohan=ko kitab **d-ii**
- Ram gave Mohan a book

- raam=ne us baat=par **zor di-yaa**
- Ram put an emphasis on that topic

- pulis=ne logon=par hamlaa ki-yaa
- ‘(The) Police attacked the people’

Vaidya et al., 2014, 2016, 2024

Annotating Eventive Nouns

- Noun phrases
  - The news **describes** the **shipping accident**.
  - He has **recovered** from a **heart attack**.
  - We **saw** a **laughing** child.
- Proper nouns
  - **The fall** of the Berlin Wall on 9 November 1989, during **the Peaceful Revolution**, **marked** **the beginning** of **the destruction** of the Berlin Wall and the figurative Iron Curtain, as East Berlin **transit restrictions** were **overwhelmed** and **discarded**. An end to **the Cold War** was **declared** at **the Malta Summit** in early December.
- Pronouns or anaphors
  - Ultimately, **one** of Lincoln's – and Grant's – greatest **achievements** had to be **winning the American Civil War** and thus **preserving** the Union.

https://en.wikipedia.org/wiki/Fall_of_the_Berlin_Wall
https://www.historyextra.com/period/victorian/abraham-lincoln-life-achievements-death/

Annotating Eventive Nouns

- Empty nouns
  - Two **incidents of sexual harassment** have been found.
- Preposition + noun
  - The diabetes drugs are currently **under development**.
- Nouns that describe states
  - When she **won**, her eyes **shone** with **happiness.**
  - The Conquest of Happiness by Bertrand Russell was **published** in 1930.

Nominal Events

1. Candidate is required to deliver his **nomination** paper on or before last date for filing of **nomination** fixed by the ECI.
2. Kamala Harris received a **nomination** from Democrats to fight against Trump.
3. Brazil official entry '*I'm Still Here*' earn a **nomination** in the Best International Feature category at the 97th Academy Awards.
4. Hence her **nomination** was rejected.
5. He lost the **election**.
6. He was thinking about filing a **nomination** for the upcoming **election**.

Knowledge acquisition

- Labeling
  - Inter annotator agreement
  - Cost and time
- Knowledge sources
  - WordNet
  - FrameNet/VerbNet
  - Wikipedia
- Domains
  - Biomedical
  - Sports/politics
- Languages

The Communications of ACM

- https://cacm.acm.org/